5th Feb 2026 | CS6370

⇒ Mean Reciprocal Rank (MRR) → tells how well a sys. ranks the correct answer near the top of its results.

$$\boxed{MRR = \frac{1}{|Q|} \sum_{i=1}^{|Q|} \frac{1}{rank_i}}$$

Spellcheck
aple → able, maple.. apple

- Evaluation → Research Methodology
  ⇒ Algo A outperforms Algo B on task C over dataset D under assumptions E across evaluation measure F.
  ⇒ Hypothesis testing

⇒ Spellcheck
  → part of pre-processing

- Key problems -
  1. how do we map variants of words to their root forms
  2. how do we overcome spelling errors?

extra quesⁿ

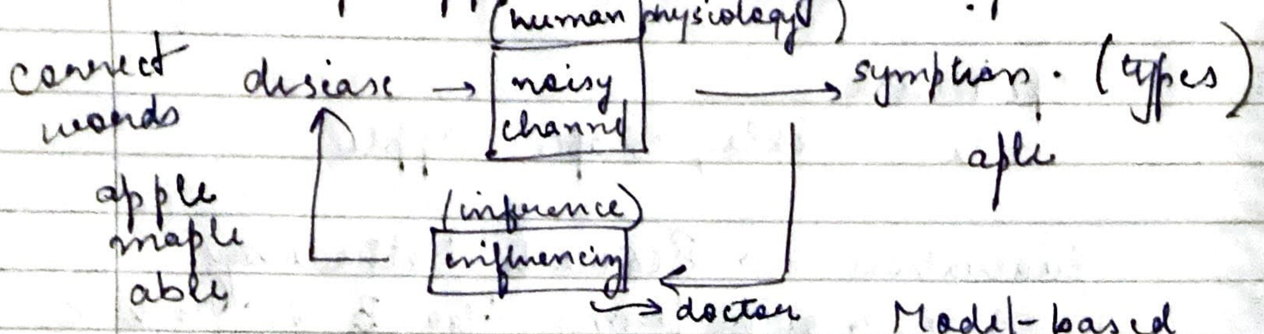
- 1. Why is correcⁿ of spelling is impt? 2. Why are spelling errors made? 3. Categories of spelling errors? 4. What factors can we exploit to recover from them?

ans.

1. human errors, OCR, STT
2. homophones, no mapping btw. stems & pronouncia (context - sensitive)
3. non words vs words / typographic / cognitive
4. edit, keyboard, pronunciaⁿ, content, syntax, source specific (like OCR)

- there are 2 views of spellcheck
  → traditional spell check based on dictionary
  → context-based (statistical / AI / grammar-based)

- bottom-up approach → diagnosis problem



Model-based & consistent-based diagnosis

- eq. of Britney Spears

- Bayesian process to solve this? → treats spelling correcⁿ as a probability problem

$$P(W|T) = \frac{P(T|W)\,P(W)}{P(T)}$$

W → word, T → typed word T

P(W) → prior
P(T|W) → likelihood
P(W|T) → posterior

- why vector spaces is not a good method to solve this? → spell-checking is more abt. exact charac. errors, while vector spaces are better at semantics (meaning), not spelling.
→ OOV problem → misspelled words usually map to <UNK> token.
  → also, leading to high computational costs